

Student Name:

HU ID #:



Habib University

EE-371/CS-330/CE-321 Computer Architecture and Organization – Spring 2024

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Time = 40 minutes

Quiz 02

Max Points: 20

Instructions:

- Smart watches, laptops, and similar electronics are strictly NOT allowed.
- Answer sheets should contain all steps, working, explanations, and assumptions.
- Attempt the quiz on clean papers with black/blue ink.
- Print your name and HU ID on all sheets.
- Turn in your question paper along with your answer sheets.
- You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

CLO Assessment:

This assignment assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	<i>Explain</i> the role of ISA in modern processors and instruction encodings and assembly language programming	
CLO 2	<i>Explain</i> the architecture and working of a single cycle processor	✓
CLO 3	<i>Design</i> the architecture to mitigate issues of a pipelined processor	
CLO 4	<i>Analyze</i> the performance of cache operations	

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Q1: [Logical instructions, procedure call, sequential multiplication]

a) Assume we have following assembly code: (3 points)

```
addi x1, x0, 0xFF
addi x2, x0, 0x55
or   x3, x1, x2
and  x4, x1, x2
xor  x5, x1, x2
```

1. Write values of x3, x4 and x5.

Sol:

$x3 = 0xFF$
 $x4 = 0x55$
 $x5 = 0xAA$

b) Write assembly language instructions for the give C code. Use same registers as variable names. Use the RISC-V procedure call convention. Remember that x5, x6 and x7 are temporary registers and do not need stack. However, return address for calling a function (*jal* & *jalr* instruction) uses x1 register, which needs stack. (3 points)

```
main(){
    x5 = x5 - 5;
    funct_1();
    return;
}

funct_1(){
    x6 = x6 - 6;
    funct_2();
    return;
}

funct_2(){
```

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```
x7 = x7 - 7;  
return;
```

```
}
```

Sol:

main:

```
addi x5, x5, -5  
jal x1, func-1
```

func-1:
addi s8, sp, -8 # Since x1 will be
over-written

```
sl x1, c(sp)  
addi x6, x6, -6  
jal x2, func-2 # x1 over-written here
```

```
ld x1, c(sp)  
addi sp, sp, +8  
jal x0, c(x1)
```

func-2:
addi x7, x7, -7
jal x0, c(x1)

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c) Multiply 1010₂ with 1101₂. Run the algorithm twice in parallel assuming that we have 0 and 11000011₂ in product at start. (4 points)

Sol:

S. No.	Multiplier(a)	Multiplicand(b)	Product (0)	Product (11000011)
0			0000 0000	1100 0011

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Q2 [Single cycle processor, datapath]

- a) Assume we have a RISC-V encoded (machine language) instruction having value $0x00C6BA23$. Use the table given below to find the value of opcode and instruction.

Name (Bit position)	Fields				
	31:25	24:20	19:15	14:12	11:7
					6:0
(a) R-type	funct7	rs2	rs1	funct3	rd
(b) I-type	immediate[11:0]		rs1	funct3	rd
(c) S-type	immed[11:5]	rs2	rs1	funct3	immed[4:0]
(d) SB-type	immed[12:10:5]	rs2	rs1	funct3	immed[4:1:11]

Sol:

instruction opcode = 0160011
From table $\Rightarrow$ S-type or sd instruction

- b) Use table below to find the values of ALU control input for the instruction.

Instruction opcode	ALUOp	Operation	Funct7 field	Funct3 field	Desired ALU action	ALU control input
ld	00	load doubleword	XXXXXXX	XXX	add	0010
sd	00	store doubleword	XXXXXXXX	XXX	add	0010
beq	01	branch if equal	XXXXXXXX	XXX	subtract	0110
R-type	10	add	0000000	000	add	0010
R-type	10	sub	0100000	000	subtract	0110
R-type	10	and	0000000	111	AND	0000
R-type	10	or	0000000	110	OR	0001

Sol:

sd means ALUOp = 00 & ALU input = 0010

- c) What is the value of PC for next instruction when this instruction executes? Assume the PC for current instruction is $0x100$.

Sol:

$PC = PC + 4$ since NOT branch
 $\Rightarrow PC_{next} = 0x104$

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- d) Show/write the control signal values for all the multiplexers when this instruction executes.

Sol:

Control signals for a sd/store instruction
See slides/book

- e) What are the two ALU operands for this instruction? (7 points for ALL parts)

Sol:

For reference: instruction is
sd x12, 20(x13)
 $\Rightarrow$ ALU operands are 20 & x13

Q3: Assume we have a special RISC-V implementation for which the instruction encoding scheme is using extra 5 bits (in addition to existing 32-bit instructions). These extra 5 bits are added at the end of an instruction to implement 3 operands add and subtract unit e.g.

add x1, x2, x3, x4 // $x1 = x2 + x3 + x4$

sub x1, x2, x3, x4 // $x1 = x2 - x3 - x4$

For this architecture, draw the register file and ALU unit with all the necessary data and control signals. Specify correct width of each signal and also interconnect these two units. (3 points)

rs3(5-bits)	funct7	rs2	rs1	funct3	rd	opcode
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Sol:

